

Group members: Amogh, Connor, Kendall

Github Repository link: <https://github.com/kdgross48/python-collaboration>

Objectives:

- Connor:
 - We are making a console-based differential equations solver. We might add a front end eventually but for now a console-based program seems enough for displaying computations. We will use pre-existing knowledge of differential equations to solve differential equations in the program. However, I will probably learn more about solving them. The goal for this project is to have users input a differential equation and have the program consider many possible solutions and display them to the user.
- Amogh:
 - Want the program to solve 1st, 2nd, and probably 3rd order differential equations.
 - I want to try adding a Laplace Transform function to solve differential equations that cannot be solved by normal methods.
 - This project's goal is to have users input the order of differential equations and then input the coefficients.
 - The program must also figure out the method of solving the differential equation, whether it be by conventional methods or the Laplace Transform.
- Kendall:
 - Our project focuses on building a differential equation solver that can handle first, second, and potentially third-order equations.
 - Users will input the order and coefficients, and the program will determine an appropriate solution method.
 - My main responsibility is to develop the approach for solving these equations, whether through analytical techniques or numerical methods.
 - I plan to explore different solving strategies to ensure accuracy and efficiency.
 - As a team, we will establish clear goals, choose the best tools for implementation, and create a structured timeline to guide our progress.
 - We will use the programming language of Python and will be creating this project in Spyder.

Select Technologies and Tools:

- Python, Spyder. The program will be run entirely on the Spyder console.

Project Timeline:

https://northeastern-my.sharepoint.com/:x:/g/personal/ng_con_northeastern_edu/Efiw_iDbP1VEhc2vHEUyB4sBsJtPS5Ld8dJVJ6DKxS_3Yw?e=CnZe4c